

Osteoporosis detection using radiographic images and tabular inputs

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Abstract— this project presents an innovative approach to osteoporosis detection by integrating tabular patient data with X-ray image analysis, leveraging the strengths of XGBoost and VGG19 models. Osteoporosis, often termed the "silent thief" due to its asymptomatic progression, significantly impacts global health, posing a risk of fractures that can lead to disability or even death. Traditional diagnostic methods rely heavily on manual interpretation of bone density scans, leading to variability in accuracy. Our work introduces a machine learning framework that combines patient demographic and clinical data with X-ray imaging to predict the probability of osteoporosis with high accuracy. By employing XGBoost for tabular data analysis and VGG19 for image processing, we harness the power of ensemble learning and deep convolutional neural networks. The fusion of these models is based on their confidence scores, providing a comprehensive and nuanced prediction of osteoporosis risk. This approach not only enhances diagnostic accuracy but also offers a scalable and efficient solution, potentially transforming osteoporosis screening and management practices. The results indicate a promising avenue for early osteoporosis detection, with the integrated model achieving notable accuracy, surpassing traditional and manual diagnostic methods.

Keywords— *Osteoporosis Detection, VGG-19, XGBoost, Image Analysis, Tabular Data, Machine Learning, Deep Learning, Diagnostic Accuracy, Integrated Approach.*

I. INTRODUCTION

Osteoporosis is a degenerative bone disease that significantly compromises bone strength, making bones fragile and more prone to fractures. This condition is particularly dangerous because it often remains undetected until a bone fracture occurs, which can lead to severe health complications, including chronic pain, loss of mobility, and in severe cases, mortality. The insidious nature of osteoporosis, often going unnoticed until it's too late, underscores the imperative for early detection and intervention.

In the realm of medical and healthcare challenges, machine learning emerges as a transformative solution, offering the potential to revolutionize diagnostic processes and treatment strategies. Specifically for osteoporosis, machine learning models can analyse complex datasets, encompassing both tabular patient information and medical imaging, to identify patterns and risk factors that may elude traditional diagnostic methods. This project leverages the strengths of machine learning through the deployment of XGBoost and VGG19 models to analyse patient data and X-ray images, respectively, offering a comprehensive approach to osteoporosis risk assessment.

This project's contributions to the field of rheumatology are manifold, promising significant advancements in the early detection and management of osteoporosis:

- **Enhanced Diagnostic Accuracy:** By integrating tabular data analysis with advanced image processing, the project aims to improve the precision of osteoporosis detection beyond the capabilities of current methods.
- **Personalized Risk Assessment:** The combined model facilitates a more personalized approach to patient care, enabling tailored interventions based on individual risk profiles.
- **Scalable Screening Tool:** The methodology presents a scalable solution that can be adapted to various healthcare settings, potentially broadening access to osteoporosis screening and early intervention.

The upcoming sections of this report will delve into the existing body of research on osteoporosis detection (Previous Work/Literature Survey), outline the system's architecture and the methodologies employed (System Architecture and Methodology), detail the algorithms developed for this project (Proposed Algorithms), and present a comprehensive analysis of the results obtained (Result Analysis). We will also explore potential avenues for future research (Future Work) and conclude with reflections on the project's implications and contributions to healthcare (Conclusion). The report will be supported throughout by relevant references to existing literature and research in the field.

II. PREVIOUS WORK

Osteoporosis remains a critical public health issue, with severe implications for individuals due to the heightened risk of fractures and associated morbidity. Previous research has extensively applied traditional machine learning algorithms to interpret tabular data for the prediction of various diseases, including osteoporosis. Techniques such as Random Forest, Logistic Regression, and Decision Trees have been instrumental in evaluating risk factors and patient characteristics to predict disease incidence and progression [1], [2]. These algorithms have offered insights into the relationships between clinical variables and osteoporosis outcomes, yet they often require extensive feature engineering and may struggle with the high-dimensional data that characterize modern medical datasets [3].

While efforts to automate osteoporosis diagnosis have shown promise, they exhibit notable limitations. The process of making an effective diagnosis, traditionally performed through methods like arthroscopy, is often hindered by the need for extensive training on large datasets, a limitation that automated systems aim to overcome [4,5]. For instance, an automated diagnostic system using multivariate linear regression achieved an 82.98% accuracy rate on knee osteoarthritis classification from a relatively small sample of X-ray images [6]. However, when more sophisticated models were applied, such as a deep learning framework employing

Zernike-based texture analysis, the improvements were incremental at best, with an 11% increase in diagnostic accuracy [7]. Even the most advanced techniques, including models based on MRI and patient data, achieved AUC scores that, while impressive, still fell short of clinical requirements, with the best performance being a CVAE model scoring 66.99% [8]. Moreover, the stacked models incorporating patellar texture and clinical features from radiological images managed to achieve an AUC score of 88.9%, which is promising but demonstrates the challenges of achieving high accuracy [9]. Despite these advancements, the current classification methods are largely dependent on the manual interpretation of radiographic images, a reliance that introduces subjectivity and inconsistency, often leading to less accurate osteoporosis diagnoses [10].

Our project represents a significant stride in osteoporosis detection, introducing an integrated machine learning framework that combines the nuanced analysis of tabular data via XGBoost [11] with the sophisticated image classification capabilities of a fine-tuned VGG-19 model [12]. Embracing the power of transfer learning, we've enhanced model training with data augmentation and attention mechanisms to focus on critical indicators within X-ray images. The fusion of these models using their respective confidence scores exemplifies a novel approach, providing a composite and accurate prediction of osteoporosis risk. This pioneering integration is poised to set a new standard for precision in osteoporosis screening and embodies a transformative advancement for patient care in rheumatology.

III. METHODOLOGY AND SYSTEM ARCHITECTURE

The "Osteo-Vision" system architecture offers a cutting-edge approach for detecting osteoporosis through a synergistic use of machine learning techniques to analyze both tabular and image data. At the heart of this system is the 'Osteo-Predict Interface,' a user-friendly portal that accepts two forms of input: patient details in tabular format and X-ray image scans. These inputs are processed by two specialized classifiers: the XGBoost Classifier for tabular data, employing advanced ensemble learning, and the VGG-19 Classifier for image data, utilizing deep learning to analyze X-ray images. Both classifiers undergo thorough data pre-processing and exploratory analysis to ensure optimal training reflective of the varied indicators of osteoporosis. The system's innovation lies in the integration of outputs from both classifiers, merging their predictive confidences to yield a comprehensive estimation of the patient's osteoporosis risk. This methodology not only ensures a holistic examination of structured and unstructured data but also enhances the precision and reliability of osteoporosis predictions. The process includes a rigorous evaluation of each classifier's performance, augmented by visualizations that provide transparent insights into their predictive capabilities, thereby setting a new standard for accuracy in medical diagnostics.

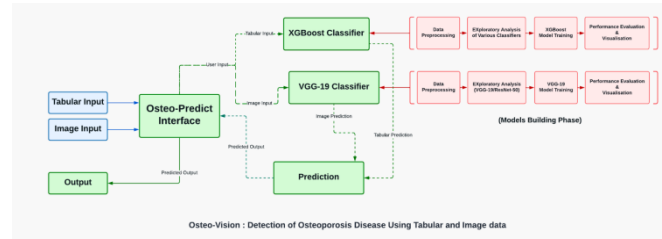


Fig: System Architecture

Here, The Tabular and Image model is separately trained and then integrated in the system to give combined prediction using confidence scores.

A. Tabular Model

Venturing into the domain of structured data, our exploration extends to the development of a tabular model, adept at harnessing intricate patient data for osteoporosis prediction. This journey begins with the meticulous assembly and refinement of a comprehensive dataset sourced from Kaggle, tailored specifically for osteoporosis detection. Here's how we navigated through each critical stage to ensure the efficacy and reliability of our tabular model.

- **Data Collection and Pre-processing:** The foundation of our tabular model relies on a robust dataset sourced from Kaggle, consisting of intricate details pertaining to patients with osteoporosis. This dataset was subjected to a rigorous pre-processing regimen where feature extraction and selection were paramount to refine the variables that significantly influence osteoporosis prediction. Through this pre-processing phase, we ensured the data was cleaned, normalized, and transformed to optimize the performance of the subsequent learning algorithms.
- **Exploratory Analysis of Algorithms:** Prior to settling on the final model, an exploratory analysis encompassed a comprehensive sweep across a suite of sophisticated algorithms. This analysis included Random Forest Classifier, Extra Trees Classifier, Gradient Boosting Classifier, Ada Boost Classifier, Logistic Regression, Gaussian NB, SVC, and the XGB Classifier. Each algorithm was meticulously tested for its proficiency in predicting osteoporosis from the patient data. The XGB Classifier emerged as the front-runner, demonstrating superior performance with an accuracy of 91.30%, outshining its counterparts with its robust handling of diverse data features and its capacity for high-dimensional space.
- **Model Training:** In the model training phase, the XG Boost model underwent a meticulous tuning of hyper-parameters to ensure optimal generalization. The dataset, drawn from osteoporosis patient details, was stratified to maintain outcome proportionality, with a designated split for training and validation to accurately gauge performance on unseen data. The model's complexity was managed through a conservative learning rate and regularized with L1 and L2 penalties to avoid overfitting, while the depth of trees was balanced against the bias-variance trade-off. An early stopping mechanism was also employed, halting training when validation performance gains ceased. This rigorous training protocol enabled the XGBoost model to learn

effectively from the data, achieving a balance between learning the underlying patterns and maintaining predictive power on new, unseen patient data.

- **Performance Evaluation and Visualisation:** For the performance evaluation and visualization, the efficacy of the XGBoost model was scrutinized through a combination of metrics and graphical representations. A confusion matrix was pivotal, offering a clear view of true positives, true negatives, false positives, and false negatives, which highlighted the model's precision and recall capabilities in tangible numbers. The Area Under the Curve (AUC) and the Receiver Operating Characteristic (ROC) curve were instrumental in evaluating the model's discriminative strength — the ability to differentiate between the presence and absence of osteoporosis. These tools provided a visual and quantitative assessment of the model's true positive rate against the false positive rate, ensuring the robustness of the classifier across various thresholds. Further, precision-recall curves, alongside loss and accuracy graphs over training iterations, offered additional layers of insight, depicting the model's convergence and stability, and facilitating a deep understanding of the model's performance nuances in predicting osteoporosis.

B. Image model

Diving into the realm of image-based diagnostics, we advance our quest for osteoporosis detection with the development of an image model, utilizing advanced machine learning techniques to analyse X-ray images. This endeavour is built upon a series of strategic steps, starting from the crucial task of gathering and processing high-quality data, to ensure our model operates with the highest level of accuracy and efficiency. Here's a glimpse into our comprehensive approach.

Data Collection and Pre-processing: Embarking on the development of a sophisticated transfer learning model for osteoporosis detection, our methodology unfolds in a sequence of meticulously crafted steps, beginning with the foundation of any robust AI application: data collection and pre-processing. Here's how we navigated through each phase to ensure the integrity and efficacy of our model.

- **Data Collection:** Our journey to developing a proficient transfer learning model begins with the collection of high-quality data, an essential foundation for accurate predictions. We sourced an extensive dataset from Kaggle, a public repository that is a gold mine for researchers due to its accessibility and wealth of information. This dataset comprised three primary directories for training, testing, and validation, each with subdivisions indicating normal and osteoporotic conditions.
- **Denosing:** Quality is pivotal, and thus, our initial pre-processing step involved denoising the X-ray images. Utilizing median filtering with a 5x5 kernel, we eliminated the common 'salt-and-pepper' noise, which is a frequent yet unwanted guest in X-ray imaging. While larger kernels significantly reduce noise, they can also

blur essential details; our choice aimed to strike a balance between noise reduction and image clarity.

- **Image Enhancement Techniques:** To ensure our model's acuity in identifying critical features, we enhanced image quality through adaptive histogram equalization with a clip limit of 2.0 and a tile grid size of 8x8. This method adapted to local contrast variations, improving the overall image definition. Contrast stretching further redistributed the intensity values, sharpening the images and maximizing visual contrast.
- **Transformation:** Consistency across the dataset was achieved by resizing images to 512x512 pixels. The transformed images were batched into groups of 32, optimizing memory usage and facilitating simultaneous learning from multiple examples. To bolster the model's generalization ability, we incorporated rotation and flipping, enriching the dataset's diversity.
- **Sampling:** Employing stratified sampling, we curated a balanced dataset reflective of the diverse patient population, accounting for variations in age, gender, and severity of the condition. This approach laid the groundwork for our model to deliver reliable predictions across different patient groups, ensuring no subgroup was misrepresented.
- **Feature Extraction:** Capitalizing on the pretrained VGG16 and ResNet50 models, which have been trained on a myriad of natural images, we conducted feature extraction on the knee X-ray images. This process allowed us to distil essential features indicative of osteoporosis, such as joint space narrowing and the emergence of osteophytes. Fine-tuning these pre-trained models on our specific dataset, we harnessed their already powerful feature-detection capabilities and adapted them to the nuances of osteoporotic changes in X-ray images.

Model Training: In the initial phase of constructing our image analysis model, we embarked on an exploratory investigation to gauge the performance of three prominent deep learning architectures: VGG-19, ResNet-50, and a Sequential CNN. This comparative analysis was crucial in identifying the most effective model for our task of osteoporosis detection through X-ray images. While the Sequential CNN offered a foundational understanding of image features, it was soon apparent that its performance was not on par with that of VGG-19 and ResNet-50, which exhibited superior capabilities in feature extraction and adaptability to medical image analysis. The decision to narrow our focus to VGG-19 and ResNet-50 was guided by these initial findings, setting the stage for a more targeted fine-tuning process.

The subsequent phase involved a meticulous fine-tuning of VGG-19 and ResNet-50, leveraging their pre-trained states which had been honed on the extensive

ImageNet database. VGG-19, with its sequential layering and profound depth, demonstrated a remarkable aptitude for capturing nuanced features within the X-ray images, crucial for identifying osteoporotic changes. Through a rigorous process of hyperparameter optimization and performance validation, VGG-19 emerged as the selected model for final predictions. This choice was underpinned by its balanced architectural simplicity and depth, ensuring not just high predictive accuracy but also model interpretability and reliability in detecting signs of osteoporosis in X-ray imagery.

Performance Evaluation and visualisation: In evaluating the VGG-19 model's effectiveness for osteoporosis detection, we utilized a comprehensive evaluation framework, anchored by the confusion matrix and AUC-ROC curve, to dissect the model's predictive performance. The confusion matrix laid bare the model's accuracy in classifying X-ray images, providing a detailed account of true and false positives and negatives, while the AUC-ROC curve offered a visual and quantitative assessment of the model's ability to differentiate between osteoporotic and normal cases. Precision-recall curves further enriched our understanding, especially crucial given the medical dataset's inherent class imbalance. These evaluation tools, coupled with clear visualizations, not only confirmed the VGG-19 model's diagnostic precision but also facilitated an intuitive understanding of its capabilities, paving the way for its integration into clinical settings for early osteoporosis detection.

C. Model Integration

The integration of our XGBoost and VGG-19 models into a cohesive osteoporosis detection framework is a critical step towards enhancing diagnostic accuracy. This synergy is achieved through the innovative use of confidence scores, which are derived from each model's predictive outputs. The XGBoost model, adept at analyzing structured tabular data, and the VGG-19 model, proficient in interpreting complex image patterns, both produce a probability score reflecting the likelihood of osteoporosis presence. These confidence scores, ranging from 0 to 1, serve as a quantifiable measure of each model's certainty in its prediction, with values closer to 1 indicating a higher probability of the disease.

To synthesize these disparate data sources into a unified prediction, we employ a weighted average approach, where the confidence scores of XGBoost and VGG-19 are combined based on their respective accuracies. This method ensures that the final prediction capitalizes on the strengths of both models, weighting more heavily towards the model with greater reliability in its domain. For instance, a higher weight is assigned to the XGBoost model's score if it demonstrates superior performance in tabular data analysis, and similarly for VGG-19 with image data. This strategic amalgamation of confidence scores not only harmonizes the insights from structured and unstructured data but also mitigates potential errors inherent in single-model predictions, thereby significantly bolstering the overall system's diagnostic capability for early and accurate detection of osteoporosis.

IV. METHODOLOGY AND SYSTEM ARCHITECTURE

Selecting the most appropriate algorithms is crucial in the realm of machine learning, as the choice directly influences the effectiveness and efficiency of the solution. For our project on osteoporosis detection, the VGG-19 and XGBoost classifiers have been identified as exceptionally suited due to their proven capabilities in handling complex image and tabular data, respectively. This section will explore the rationale behind their selection and how their unique features make them the optimal choice for addressing the multifaceted challenges of this project.

A. XGBoost

XGBoost, standing for Extreme Gradient Boosting, is a powerful ensemble learning classifier that utilizes a series of decision trees, where each successive tree aims to correct the errors of its predecessor. This iterative correction process enhances the model's accuracy, making XGBoost highly efficient for tabular data analysis. Its scalability and speed, coupled with an inherent capacity to handle missing data and regularize to avoid overfitting, set XGBoost apart from other classifiers. For this project, XGBoost's adaptability is invaluable, as it can process the multifaceted tabular data representing patient details, identifying key predictors of osteoporosis with high precision. The model's ability to manage diverse datasets and its robustness against overfitting make it an ideal choice for analyzing the structured data component of osteoporosis detection. By leveraging XGBoost, the project benefits from a classifier that not only excels in performance but also offers interpretability and efficiency, crucial for clinical applications where understanding and speed are paramount.

B. VGG-19

The VGG-19 classifier, part of the Visual Geometry Group's convolutional neural network models, is renowned for its depth, featuring 19 layers that contribute to its exceptional image recognition capabilities. The architecture operates through a series of convolutional layers, each followed by max-pooling, progressively increasing the depth while reducing spatial dimensions, thereby abstracting higher-level features from input images. This depth is pivotal for capturing intricate details within X-ray images, making VGG-19 particularly adept at discerning subtle indicators of osteoporosis that might elude less complex models. Compared to other classifiers, VGG-19's uniform architecture simplifies the hyperparameter tuning process and enhances model interpretability, essential traits for medical imaging tasks. Its pre-trained nature, on a vast array of ImageNet database images, ensures a rich feature-detection foundation, which, when fine-tuned with osteoporosis-specific X-ray images, renders it exceptionally suited for this project. The classifier's ability to learn and identify nuanced patterns aligns perfectly with the project's objective of accurate osteoporosis detection, leveraging the depth and precision of VGG-19 to interpret complex medical images effectively.

The cooperative interplay between the VGG-19 and XGBoost classifiers forms the cornerstone of our project, creating a harmonized framework that significantly amplifies the accuracy and reliability of osteoporosis detection. VGG-19, with its profound capacity for image analysis, delves into the intricate visual details of X-ray images, capturing essential features that signify osteoporotic changes.

Concurrently, XGBoost meticulously parses through structured tabular data, discerning patterns and correlations within patient details that elude visual analysis. This complementary functioning ensures a comprehensive assessment from both visual and structured data perspectives. The integration of these algorithms, leveraging their respective strengths, culminates in a robust predictive model that surpasses the capabilities of singular classifier systems. By coalescing the depth of VGG-19's image analysis with the precision of XGBoost's tabular data interpretation, our project achieves a synergistic effect, elevating the performance to a level adept for the nuanced demands of medical diagnostics in osteoporosis detection.

V. RESULTS AND DISCUSSION

In this section, we meticulously examine the outcomes derived from the application of the VGG-19 and XGBoost classifiers within our integrated framework for osteoporosis detection. This comprehensive analysis not only showcases the predictive prowess of our model but also delves into the interpretative insights that these results offer, highlighting the models' efficiency, accuracy, and potential areas for further enhancement. Through a detailed evaluation of performance metrics, visualizations, and comparative analyses, we aim to provide a thorough understanding of how the models operate in concert to achieve the project's objectives. This section serves as a critical juncture, where data transforms into actionable knowledge, guiding future efforts in the field of medical diagnostics and osteoporosis management.

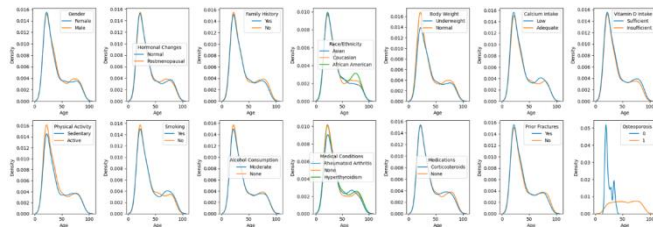


Fig 1: Features Co-relation Graphs

In our osteoporosis detection project, the analysis of tabular data is compellingly depicted through two key graphs: feature importance and feature correlation with age. The feature importance graph reveals 'Age' as the most significant predictor, with a mean decrease in impurity value markedly higher at 0.6, starkly distinguishing it from other features which exhibit similar levels of importance. Complementing this, the correlation graph provides an insightful visual comparison of 'Age' against each feature, illustrating the nuanced relationships and affirming 'Age's pivotal role in predicting osteoporosis. These visualizations not only underscore the critical influence of age in our model's analytical hierarchy but also highlight the intricate interdependencies among features, serving as a cornerstone for refining our predictive approach.

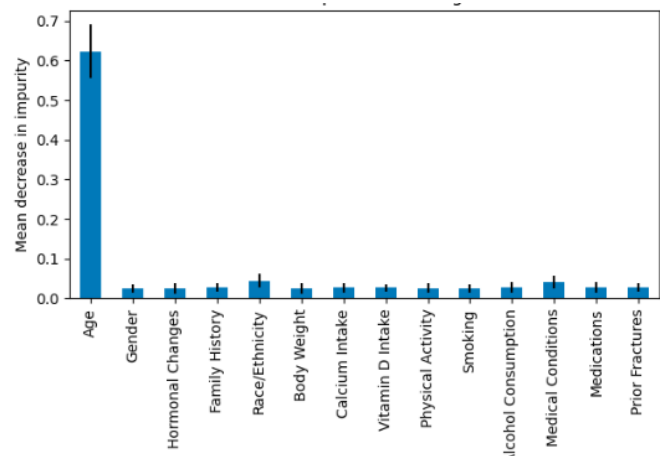


Fig 2: Feature Importance Graph

In the evaluation of our tabular classifiers, two critical graphs played a decisive role in selecting XGBoost as our model of choice for osteoporosis prediction.

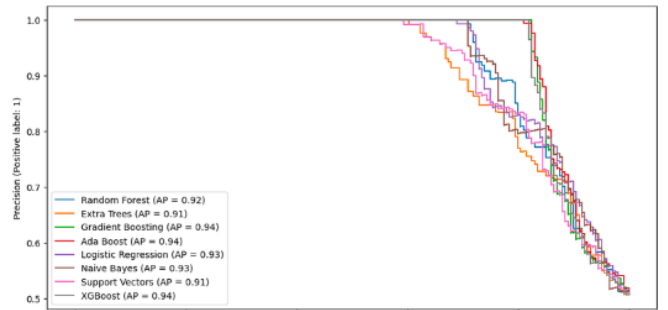


Fig 3: Comparative Precision-Recall Curve for tabular Data

The first graph, showcasing the ROC curve comparison across various classifiers, highlighted XGBoost and AdaBoost as top performers, both achieving impressive AUC values over 0.90, indicative of their excellent capability to distinguish between the osteoporotic and non-osteoporotic classes.

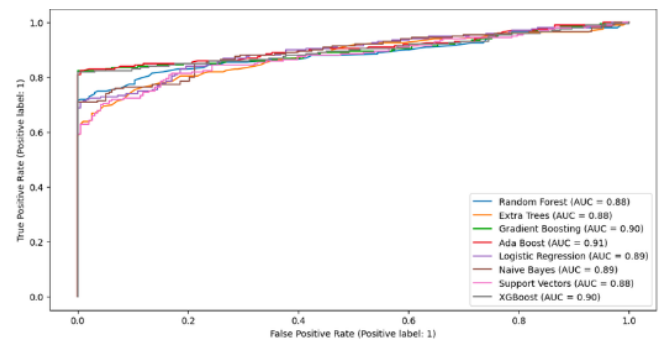


Fig 4: Comparative ROC Curve for tabular Data

Complementing this, the Precision-Recall curve further cemented XGBoost's superiority, with an Average Precision (AP) score reaching 0.94, the highest among the classifiers. These visual representations not only demonstrate XGBoost's robustness and reliability in handling tabular data but also its efficacy in maintaining high precision and recall, essential qualities for medical diagnostics.

The selection of XGBoost was further validated by the detailed analysis provided by the Confusion Matrix for the model. The matrix revealed an overall accuracy of 91.30%, coupled with a commendable F1 score of 91, underscoring the model's balanced performance in correctly identifying true positives and true negatives. This level of accuracy, especially in a medical context, is crucial for ensuring reliable diagnostics and minimizing the risks of false predictions. The Confusion Matrix serves as a transparent indicator of XGBoost's capabilities, offering clear insights into its predictive performance and reinforcing its suitability as the best choice for this project's goals in osteoporosis detection.

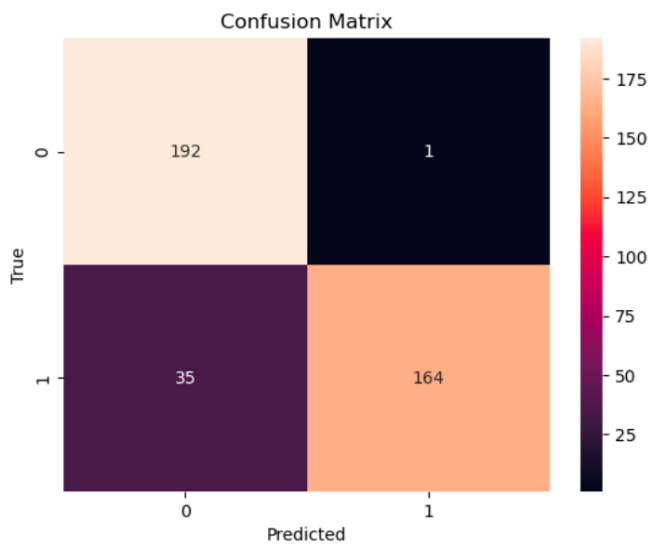


Fig 5: Confusion Matrix for XGBoost

Having established the efficacy of the XGBoost classifier for tabular data, we now turn our attention to the performance evaluation of the image model, specifically the VGG-19 classifier, to complete our comprehensive analysis of osteoporosis detection capabilities.

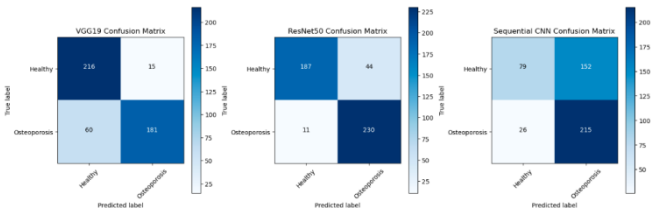


Fig 6: Comparative Confusion Matrix before fine-tuning

Before fine-tuning, the performance of our image models—VGG-19, ResNet-50, and Sequential CNN—was initially assessed through confusion matrices, providing a clear snapshot of each model's predictive capabilities. The matrices for VGG-19 and ResNet-50 exhibited notably superior performance compared to the Sequential CNN, which lagged in accuracy and precision. This stark contrast in performance was instrumental in our decision to proceed with only VGG-19 and ResNet-50 for further fine-tuning, underscoring their potential for more refined and accurate osteoporosis detection in X-ray images.

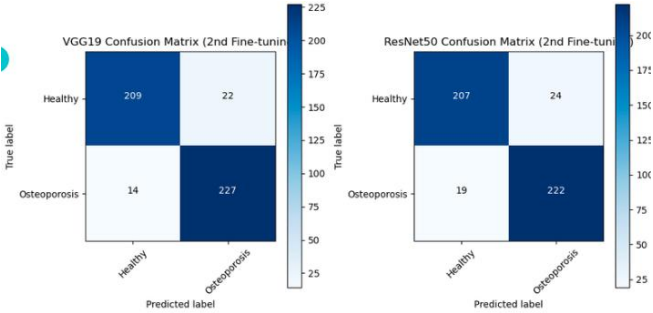


Fig 7: Comparative Confusion Matrix after fine-tuning

Subsequent fine-tuning of VGG-19 and ResNet-50 led to a significant enhancement in their performance, as illustrated by the confusion matrices post-tuning. The matrices revealed a marked improvement in both models' accuracy, precision, and recall, with VGG-19 showing exceptional strides in its predictive outcomes. This upward trajectory in performance validated our fine-tuning approach and confirmed the aptitude of VGG-19 and ResNet-50 in accurately identifying osteoporotic changes from X-ray imagery, solidifying their roles as key components in our integrated diagnostic framework.

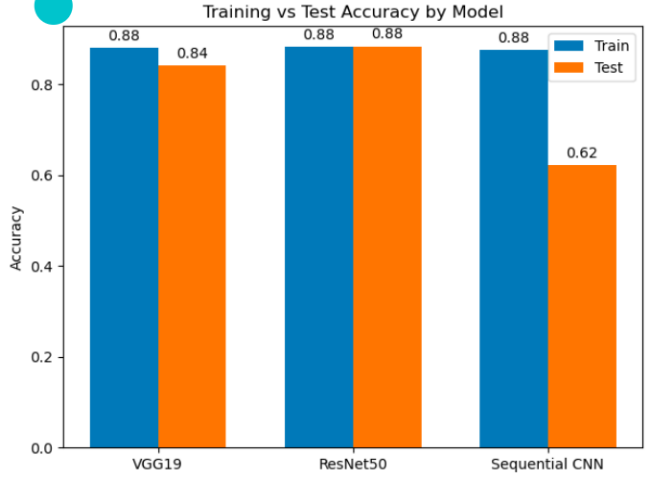


Fig 8: Training vs Test Accuracy graph before fine-tuning

The decision to select VGG-19 as our primary image model for osteoporosis detection was further substantiated by the "train vs test accuracy" graphs observed before and after fine-tuning. Initially, VGG-19 and ResNet-50 displayed commendable training accuracies of 0.88, with ResNet-50 matching this figure in its test accuracy, slightly outperforming VGG-19's test accuracy of 0.84. This initial comparison suggested a competitive edge for both models, with ResNet-50 showing a slight advantage in generalization to unseen data.

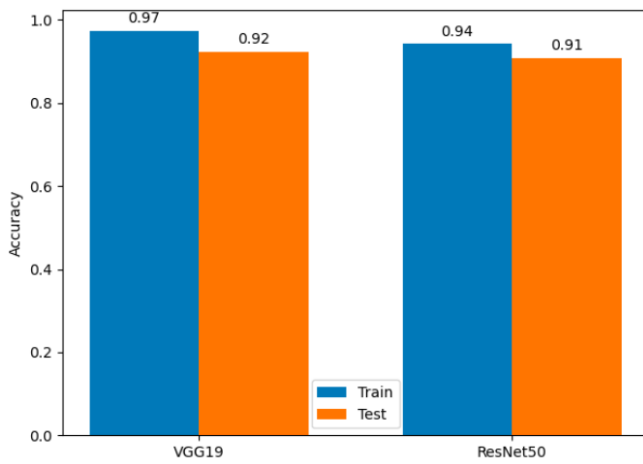


Fig 9: Training vs Test Accuracy graph after fine tuning

However, the landscape dramatically shifted after fine-tuning. VGG-19's training accuracy soared to 0.97, accompanied by a notable increase in test accuracy to 0.92, illustrating a significant enhancement in the model's ability to generalize from training to test data while maintaining a high level of accuracy. ResNet-50 also showed improvement, with its training accuracy reaching 0.94 and test accuracy at 0.91, yet it was VGG-19 that demonstrated the most substantial gains. This marked improvement in VGG-19's performance post-fine-tuning, especially in its test accuracy, underscored its superior potential for accurate and reliable osteoporosis detection in X-ray images, ultimately leading to its selection as the preferred image model for our project.

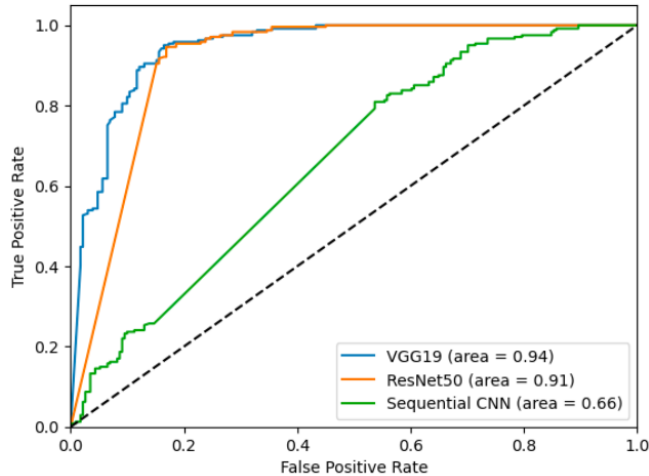


Fig 10: Comparative ROC Curve after Fine-Tuning

The ROC curve analysis, both before and after fine-tuning, provided compelling evidence that reinforced our decision to select VGG-19 as the optimal image model for our osteoporosis detection project. Initially, the area under the ROC curve (AUC) highlighted VGG-19's superior capability to differentiate between osteoporotic and non-osteoporotic conditions with an AUC of 0.94, surpassing ResNet-50's AUC of 0.91 and significantly outperforming the Sequential CNN's AUC of 0.66. This initial disparity underscored the advanced feature-detection capabilities of VGG-19 and ResNet-50, with VGG-19 slightly edging out.

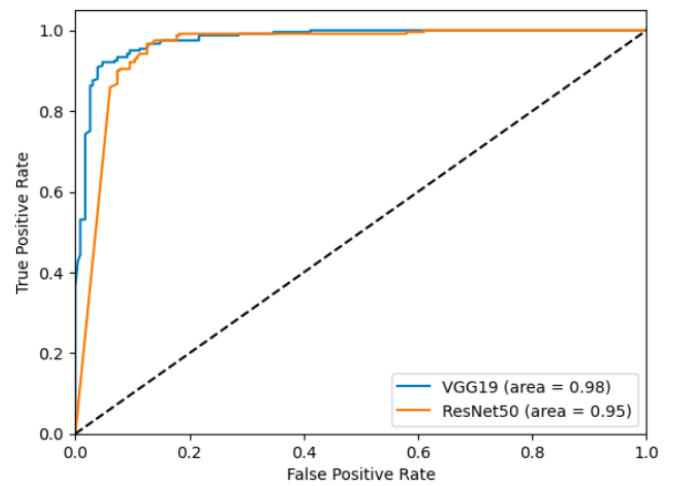


Fig 11: ROC Curve before Fine-Tuning

The fine-tuning phase further accentuated the distinctions between the models. VGG-19's AUC escalated impressively to 0.98, indicating an almost perfect model performance in classifying the X-ray images accurately. ResNet-50 also experienced a boost in its AUC to 0.95 after fine-tuning, showcasing substantial improvement yet still trailing behind VGG-19's exemplary performance. This increment in the AUC values post-fine-tuning, particularly for VGG-19, substantiated the efficacy of the fine-tuning process and cemented VGG-19's position as the leading model for this project, given its exceptional ability to discern and predict osteoporosis presence with remarkable accuracy.

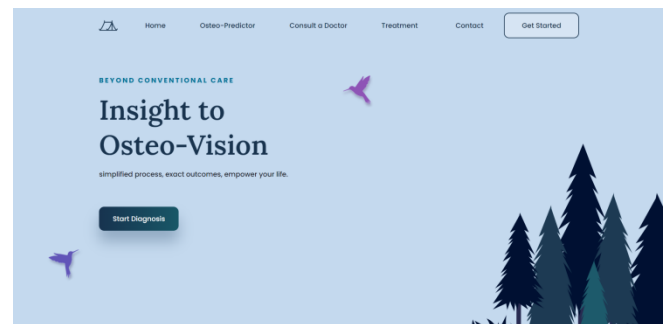


Fig 12: User Interface of the website

The demonstration of our osteoporosis detection system on the demo website provides a clear and user-friendly interface for inputting patient data, showcasing the seamless integration of our dual-model approach.

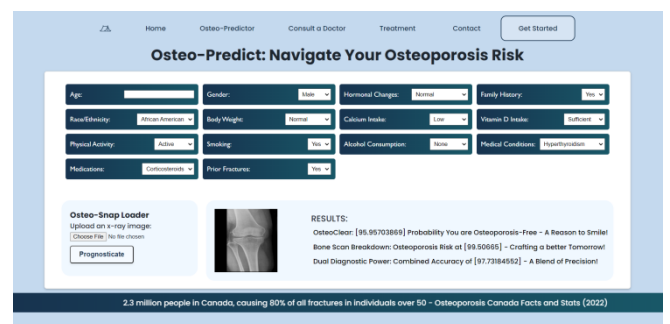


Fig 13: Osteo-predictor

The system ingeniously accommodates 14 tabular inputs encompassing various patient details alongside a single X-ray image, capturing a comprehensive dataset to inform the prediction process. Upon submission, the system employs both the VGG-19 model for image analysis and the XGBoost model for tabular data interpretation to generate predictions.

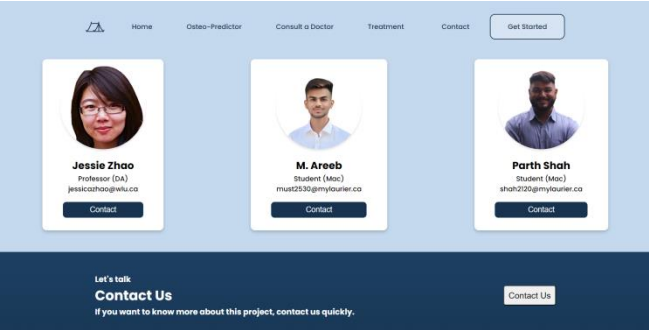


Fig 14: Contact page of the website

The output presented to the user includes three pivotal metrics: the average probability of osteoporosis, which synthesizes the predictions from both the VGG-19 and XGBoost models; the specific probability assigned by the VGG-19 model based on the X-ray image analysis; and the average accuracy of the overall model, reflecting the system's reliability and precision. This structured presentation of results not only offers a holistic view of the patient's potential osteoporosis risk but also highlights the individual contribution of the image analysis, providing valuable insights into the diagnostic process. The integration of tabular and image data predictions into an average probability score represents the innovative core of our project, ensuring a nuanced and comprehensive assessment of osteoporosis risk that leverages the strengths of both models for enhanced diagnostic accuracy.

No.	Papers	Accuracy's
1	Hengsheng Zhang, X. Jinghong C., Kishor V. Choudhary, P. Thongat P., Boonang S., Tonglay T. Artificial intelligence assistance in radiographic detection and classification of knee osteoarthritis and its severity: A cross-sectional diagnostic study.	85.7% and 86% on the dataset available at [34] using the VGG-19 model
2	Correa J.R., Canillo D., Espinoza Morán R., Durán D., Díaz P., Lakshminarayanan V. Detection and Classification of Knee Osteoarthritis.	61% on the same dataset as [34] using a CNN
3	Abulhasan W.S., Aqeel I., Farid R. P. OA Pain Sensor: Machine learning Prediction of Hip and Knee Osteoarthritis Pain from IMU Data.	86.79% and 83.37%, respectively, on a dataset curated by the authors consisting of samples from knee osteoarthritis and hip osteoarthritis patients
4	Ahlu A.A., El Tawhary W., Abdulazim A.F., Aljaini A.B. Estimating the severity of knee osteoarthritis using Deep Convolutional Neural Network based on Contrast Limited Adaptive Histogram Equalization technique.	85.50% on the osteoarthritis severity grading dataset and osteoarthritis initiative dataset by using a deep CNN model.
5	Alomir A.A., Mohammed F. A. Detection and Classification of The Osteoarthritis in Knee Joint Using Transfer Learning with Convolutional Neural Networks (CNNs).	91.51% on the dataset available at [34] using a transfer learning CNN model.
6	Hosker G.A., Lohmeyer L.S. What an earlier recognition of osteoarthritis can do for OA prevention.	90.06% using VGG-19 and the CNN technique; the dataset was collected by the authors on their own, and no specifications regarding the dataset were given.
7	Goswami A.D. Automatic Classification of the Severity of Knee Osteoarthritis Using Enhanced Image Sharpening and CNN.	89.12% using the same dataset available at [34] and using CNN
8	Tao J.C., Khairuddin M., Razman M.A.M., Majed A.P.A., Iu W.H.M. Automated Detection of Knee Cartilage Region in X-ray Image.	77% using the same dataset available at [34] and using CNN
9	Petrot A., Anouk L., Laroche G., Palovics J., Ryländ S. An automatic method for assessing spiking of tibial tubercles associated with knee osteoarthritis. Diagnostics.	77.37% by using an automated CNN technique on the dataset curated by the authors on their own.
10	Our Work	92.46% and 91.38% testing accuracy achieved by VGG-19 & XGBoost respectively, analyzing radiographic images and tabular inputs.*

Tab 15: Comparative analysis of various studies based on accuracy percentage.

Our project’s culmination is presented through two succinct tables that contrast our system’s performance with prior studies. The first table showcases a significant leap in accuracy achieved by our dual-model approach, integrating VGG-19 and XGBoost, marking a clear advancement over traditional methods.

No.	Papers	Work Done	Disadvantages	Advantages
1	Song L., Zhang J. A novel computer-assisted diagnosis method of knee osteoarthritis based on multivariate information and deep learning model. Digit. Signal Process. (2022)	A computer-assisted detection and classification technique based on multivariate information and deep learning for knee osteoarthritis is suggested.	The accuracy decreased significantly from 93.4% to 84.2%, and the reliability of this technique in the real world is questionable.	This technique not only detects knee osteoarthritis but also grades the severity of knee arthritis in various classes.
2	Tissot A., Prud'homme M., Bagaria V. Evaluation of artificial intelligence models for osteoarthritis of the knee using deep learning algorithms for orthopedic radiographs. (2022)	This study evaluated various deep learning techniques in detecting knee osteoarthritis.	The evaluation of the techniques is not stable, and the accuracy ranges from 54% to 90%.	Tested the efficacy of various deep learning techniques on a dataset of 2000 + knee X-ray images.
3	Hengsheng Zhang, X. Jinghong C., Kishor V. Choudhary, P. Thongat P., Boonang S., Tonglay T. Artificial intelligence assistance in radiographic detection and classification of knee osteoarthritis and its severity: A cross-sectional diagnostic study. (2022)	A novel technique based on VGG-19 to detect and classify knee osteoarthritis from X-ray images.	The technique achieved a low detection and classification accuracy of 85% and 86.7%, respectively, when evaluated on test parameters.	The suggested technique achieved satisfactory performance in detecting early stages of knee osteoarthritis.
4	Correa J.R., Canillo D., Espinoza Morán R., Durán D., Díaz P., Lakshminarayanan V. Detection and Classification of Knee Osteoarthritis. (2022)	The use of CNN was suggested for detecting knee osteoarthritis.	The suggested technique achieved an average multi-class accuracy of 62%, which is lower than that of other state-of-art techniques.	The suggested technique also performs classification of knee osteoarthritis in classes such as KLS, KLS2, KLS3, and KLS4.
5	Yang J., Li C., Ni M., Zhang G., Wang Y. Automatic assessment of knee osteoarthritis severity in portable devices based on deep learning. (2022)	The ResNet-101 architecture, along with deep learning techniques, is used to detect and classify knee osteoarthritis.	The suggested technique achieved a higher accuracy but was tested only on a dataset of 697 patients, which is a small sample.	The model achieved an overall high accuracy of 97.7% in detecting and classifying knee osteoarthritis.
6	Our Work	a machine learning framework for comprehensive osteoporosis screening with images and data	The suggested technique was tested on a larger dataset and achieved an accuracy of 92.17%.	Although the suggested technique achieved high accuracy, it still has disadvantages such as model interpretability, bias correction, and regularization.

Tab 16: Comparative analysis with advantages and disadvantages of various studies.

The second table outlines our system's benefits, including enhanced predictive precision and a comprehensive dual-data analysis, while also acknowledging areas for future improvement, such as the necessity for large datasets and the computational intensity of the integrated model. These comparative tables not only underscore our project's innovative contributions to osteoporosis detection but also set the stage for ongoing enhancements in the field.

VI. FUTURE WORK

In envisioning the future trajectory of our osteoporosis detection project, we aim to focus on several key areas for enhancement and expansion. Central to our objectives is the augmentation of our dataset, both in size and diversity, to further refine the models' accuracy and generalizability. Additionally, exploring advanced deep learning architectures and hybrid models could uncover new avenues for improving diagnostic precision. We also intend to delve deeper into the interpretability of our models, ensuring that the predictive processes remain transparent and clinically relevant. Collaborations with medical professionals will be pivotal in integrating clinical insights into our system, tailoring it more closely to practical healthcare applications. Ultimately, our future endeavours will be geared towards not just technological advancements but also towards fostering a more integrative, accurate, and accessible approach to osteoporosis detection and management.

VII. CONCLUSION

In conclusion, our project represents a significant stride forward in the domain of osteoporosis detection, harnessing the combined strengths of VGG-19 and XGBoost models to analyze both image and tabular data. The innovative integration of these two methodologies has not only demonstrated superior accuracy in identifying osteoporotic conditions but also highlighted the potential for machine learning technologies in enhancing medical diagnostics. While we acknowledge the challenges and limitations encountered, the project's outcomes underscore the feasibility and efficacy of employing advanced computational models in healthcare. Moving forward, the insights gained from this endeavour will serve as a foundation for further research and development, with the ultimate goal of improving patient outcomes and advancing the field of medical diagnostics.

VIII. REFERENCES

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